

Determinants of International Track-and-Field Recruitment to NCAA Division I: Wealth at the Extensive Margin, Institutions at the Intensive

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Abstract

We construct a country-level panel of international student-athletes in U.S. NCAA Division I rosters for the 2024–25 / 2025–26 season. Athlete-level records are scraped from athletic-department websites for soccer ($n = 660$) and track and field / cross country ($n = 18,657$), then hometowns are mapped to ISO-3 country codes and merged to World Bank, UCDP/PRIO, EM-DAT, UNHCR, and WGI panels. 96% of international athletes in the sample compete in track, so we characterise the empirical setting as the international track-and-field recruitment market. A two-margin decomposition is the central finding. **On the extensive margin** (whether a country sends any athletes), log GDP per capita and log population are strong predictors ($\hat{\beta} = 0.52$ and 0.18 respectively, both $p < 0.01$ in a probit on 214 countries); political stability is not significant. **On the intensive margin** (athletes per million 15–24-year-olds, among the 102 sender countries), log GDP per capita is fragile across specifications while political stability is consistently large and robust ($\hat{\beta} \approx 1.2$ in the OLS rate model, $+1.5$ when estimated cleanly on senders alone). Disaster-event counts reduce per-cohort supply ($\hat{\beta} = -0.75$, $p < 0.01$); this survives dropping the top-5 disaster-affected countries. An earlier finding that refugee outflows positively predict athlete supply is a denominator artefact: on raw counts the coefficient flips sign and loses significance. Across all specifications, R^2 in the rate model sits at 0.45–0.48, leaving more than half of cross-country variation unexplained by macro covariates—we attribute the residual to country-specific recruitment pipelines (Kenya-Iten, Jamaica-MVP, the Caribbean sprint network) that the macro variables cannot capture.

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools, included for completeness) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). The sample is dominantly track and field: only 68 of 1,830 international athletes compete in soccer. We therefore present headline results on the combined sample and report a soccer-only specification in Section ?? but characterise the empirical setting throughout as the *international track-and-field recruitment market*.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian

provinces, World Bank country names, and a manual alias map for common roster forms (“U.K.”, “B.C.”, “Czech Republic”). The mapping resolves 87.6% of track and 99.8% of soccer hometowns. Of 19,317 total athletes, **1,830 are international** (country code other than USA), and 1,819 of those merge cleanly to the 2023 World Bank macro panel.

Country covariates are taken from World Bank WDI (GDP per capita PPP, population, 15–24-year-old cohort 2018–2022 average, unemployment, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO armed-conflict data, EM-DAT natural disasters (both event counts and total deaths), the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics (refugees and asylum seekers by country of origin). For each country we use both 2023 cross-sections and 2010–2023 cumulative aggregates.

The country panel used for intensive-margin analysis contains the **102 countries with at least one international athlete in our sample**. For the extensive-margin selection equation we use a wider universe of 214 countries with non-missing macro covariates, of which 107 send at least one athlete.

A note on what we did *not* do: an initial project plan proposed a principal-component-based “Economic Distance Index” combining GDP, PPP, unemployment, and exchange-rate variables. We ultimately preferred reporting the individual covariates because their separate identification (GDP vs. political stability vs. disasters) is the substantive question and a single index would have obscured the two-margin structure that we report below.

Table 1. Descriptive statistics by income tier

Countries are split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stab
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

Per-cohort rate rises sharply with income (16× from Q1 to Q4), while **aggregate counts do not** (1.5× because Q1 contains populous countries with the well-known Kenya / Nigeria / Jamaica pipelines). The two-margin story below resolves this tension: GDP governs whether a country participates at any appreciable scale, while political stability governs the per-capita intensity of participation.

2 Specification

We decompose the country-level supply of international NCAA athletes into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c), \quad (1)$$

estimated by probit on the universe of 214 countries with non-missing covariates. The intensive margin is the per-cohort rate among the 102 sender countries,

$$\log\left(\frac{\text{athletes}_c}{\text{pop}_{15-24,c}}\right) = \alpha + \beta_1 \log \text{GDPpc}_c + \beta_2 \text{polstab}_c + \beta_3 \log(1 + \text{disaster events}_c) + \mathbf{x}'_c \boldsymbol{\delta} + \varepsilon_c. \quad (2)$$

We report Huber–White (HC1) heteroskedasticity-robust standard errors throughout. For the intensive margin we also fit Poisson and Negative Binomial models on raw athlete counts with $\log \text{pop}_{15-24,c}$ as offset.

3 Results

3.1 The two margins of athlete supply

Table 2 reports the joint two-margin decomposition. The contrast is the central finding of the paper.

	Extensive (probit) Pr(sends any athletes)	Intensive (OLS rate) log(ath./cohort)
log GDP per capita PPP	+0.52*** (0.11)	+0.24 (0.24)
log population	+0.18*** (0.06)	—
Political stability (WGI)	+0.19 (0.17)	+1.51*** (0.33)
<i>n</i>	191	102
Pseudo- R^2	0.31	0.40

HC1 robust SEs in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Extensive margin. GDP per capita and population strongly predict whether a country appears in NCAA rosters at all. The probit average marginal effect of $\log \text{GDPpc}$ on $\text{Pr}(\text{sends})$ is approximately +0.20 per log-unit at the sample mean: a country one log-unit richer ($\approx 2.7\times$) is 20 percentage points more likely to send any athletes. Political stability does not move the entry decision: in the universe of 214 countries, conditional on income and size, governance does not separate senders from non-senders.

Intensive margin. Conditional on participating, the picture inverts. The log-GDP-per-capita coefficient is statistically indistinguishable from zero; political stability is large and precisely estimated (+1.51, $p < 0.01$). A one-unit increase on the WGI political-stability index (a one-standard-deviation move across countries) is associated with $\exp(1.51) \approx 4.5\times$ more athletes per million 15–24-year-olds.

The substantive read: **wealth governs entry into the pipeline; conditional on entry, institutional quality governs intensity**. The original cross-section regressions in Section ?? pooled both margins, which gave a diluted “GDP matters, but stability matters more” framing. The two-margin decomposition is the cleaner statement.

3.2 Intensive-margin robustness

We focus on the intensive-margin (rate) model in Equation ?. Table 3 reports a battery of specifications.

Coefficient	OLS rate (total pop)	OLS rate, 15–24 cohort	Disaster events	Poisson (cohort offset)	Negative Binomial
log GDPpc	+0.22	+0.39*	+0.25	+0.07	−0.28
Political stability	+1.16***	+1.19***	+1.13***	+1.09***	+1.29***
Econ. shocks 2010–23	+0.03	+0.05	−0.10	−0.02	−0.22**
log(1 + disaster deaths)	−0.25***	−0.23***	—	−0.19***	−0.21***
log(1 + disaster events)	—	—	−0.75***	—	—
n	100	100	100	100	100
R^2 / pseudo- R^2	0.45	0.48	0.50	—	—

HC1 robust SEs throughout. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Political stability is robust. The coefficient ranges from +1.09 to +1.29 across the five specifications and is significant at $p < 0.01$ in every one. Dropping the top-20 sender countries leaves the coefficient at +0.94 ($p < 0.01$), so it is not driven by the well-known pipelines.

Disaster events is the cleaner shock measure. The headline disaster variable in earlier drafts was $\log(1 + \text{disaster deaths}_{2010-23})$. We were concerned that deaths might be dominated by a few catastrophic events. The top-5 disaster-death countries in our sample are in fact Russia, Turkey, Italy, India, and Japan (driven by the 2011 Tohoku earthquake, 2010–11 Russian heatwave, and Italian/Indian earthquakes)—all substantial NCAA senders. Dropping these five leaves the deaths coefficient at -0.20 ($p < 0.01$), so the result survives outlier trimming. Replacing deaths with $\log(1 + \text{disaster events})$, which is less sensitive to a few mega-events, yields a markedly *larger* coefficient of -0.75 ($p < 0.01$). We prefer the events specification in subsequent tables as the more conservative test: a doubling of disaster events 2010–2023 is associated with $2^{-0.75} \approx 40\%$ fewer athletes per young person.

Econ-shock count significant only under count-data distributional assumptions. In the Negative Binomial specification the count of macroeconomic shocks 2010–2023 enters at -0.22 ($p < 0.05$); in the OLS rate models it is statistically zero. The NegBin estimate likely picks up small-count countries where one extra shock matters disproportionately; we do not lean on it.

3.3 Sport-specific patterns

The international concentration is in track. The track-only panel ($n = 99$) replicates the full-sample result: $\log \text{GDPpc} + 0.35$ ($p < 0.05$), $\log \text{pop} + 0.23$ ($p < 0.05$), political stability +0.31 (n.s. in the count model with population controlled; +1.21 in the rate model). The soccer panel ($n = 24$ international-sender countries) is too small for clean identification; only $\log \text{pop}$ approaches conventional significance. A separate research design—possibly modelling school-level recruiting networks rather than country-year macroeconomics—is required for international soccer. For the rest of the paper we treat the empirical setting as the international track-and-field market.

3.4 Refugees: a denominator artefact, not a finding

A preliminary version of this paper reported a positive coefficient on refugee outflows per cohort in the rate model ($\beta = +0.77$, $p < 0.10$). On further inspection this is a denominator artefact. Re-running the same model on raw athlete counts—with $\log \text{pop}$ as a control and raw $\log(1 + \text{refugees})$ on the right-hand side instead of refugees per cohort—flips the coefficient sign:

	Per-cohort rate (orig.)	Raw count (revised)
log refugees variable	+0.74 (0.46)	− 0.13 * (0.08)
log GDP per capita	+0.40* (0.23)	+0.17 (0.18)
log population	—	+0.28*** (0.11)
Political stability	+1.21*** (0.29)	+0.12 (0.29)

The mechanism is mechanical: refugee-producing countries (Syria, Afghanistan, Venezuela, South Sudan) have war-disrupted youth cohorts, so dividing their athlete count by the small 15–24-cohort denominator inflates the rate ratio. Once we move to raw counts with population controlled, refugee outflows enter with a negative sign consistent with the disaster-events finding—countries with more forced migration produce fewer athletes overall, not more. We therefore retract the positive refugee finding and report this null / weakly negative result.

3.5 Other shocks add no independent information

We tested inflation (WDI FP.CPI.TOTL.ZG, 2010–2023 mean) and merchandise terms-of-trade change and shock-count (WDI TT.PRIMRCH.XD.WD). All four coefficients enter at conventional standard errors with point estimates indistinguishable from zero (log mean inflation −0.20; ToT change +0.14; ToT shock count −0.06). Political stability and disaster events are unchanged. The **null result** on terms-of-trade variables is itself substantive: external-price shocks do not appear to be an independent channel for international NCAA recruitment after GDP and political stability are accounted for.

3.6 Shift-share decomposition does not change the story

For completeness, we constructed a Bartik-style shift-share exposure using country shares across nine track-event groups (Sprints, Mid-Distance, Distance, XC, Hurdles, Jumps, Pole Vault, Throws, Multi-Events) parsed from roster position fields and leave-one-out aggregate event flows, $B_c = \sum_e s_{c,e} \log(1 + \Delta_e^{-c})$. The exercise has two interpretive limitations: the first-stage F -statistic on B_c as an instrument for log GDPpc is 13.5, below the $F > 104.7$ threshold for valid t -tests at the 5% level recommended by ?; and the cross-section provides too few event-mix degrees of freedom to identify a shift-share LATE. We do not lean on the IV. The single useful result is that when B_c enters the cohort-rate equation as a covariate, political stability remains +1.30 ($p < 0.01$)—this rules out the alternative explanation that the stability finding is a hidden sport-mix effect (e.g. stable European countries doing more throwing). The Bartik construction is reported in `output/bartik_results.txt`; we do not develop it further here.

4 Discussion

The empirical results support a two-margin interpretation of international NCAA track-and-field supply.

The extensive margin is governed by economic capacity and population scale. Whether a country sends any athletes at all is predicted by GDP per capita and population; political stability is not significant on this margin. The mechanism is intuitive: the fixed costs of producing one US-recruitable athlete (developmental infrastructure of *any* kind, parental income to fund coaching,

transport and equipment costs, the discovery cost on the US side) require some minimum wealth and population base. Below that threshold, no athletes are produced, regardless of the country’s governance.

The intensive margin, conditional on participation, is governed by institutional quality.

Among the 102 sender countries, political stability is the only macro variable with a consistently large and significant coefficient across OLS, Poisson, Negative Binomial, and Bartik-controlled specifications. We interpret this as evidence that the marginal pathway from a country to a U.S. NCAA roster runs through institutions—youth-development clubs, school sports infrastructure, athletic federations—rather than through pure economic push or pull. Stable states can sustain the multi-year investments that produce internationally competitive athletes; unstable ones cannot.

Disasters reduce per-cohort supply, with the effect more robust when measured as event counts than as deaths ($\hat{\beta} = -0.75$, $p < 0.01$, with deaths-based estimates of -0.20 to -0.25 across specifications). The events estimate likely captures disruption of youth sports infrastructure that does not require catastrophic loss of life to register.

Counter to a naive “economic distance” framing, GDP ratios and terms-of-trade shocks add no independent intensive-margin information once political stability is in the model. The Kenyan distance-running and Jamaican sprint pipelines are visible as outliers in our scatter plots but do not drive the panel-level result: dropping the top-20 senders leaves the political-stability coefficient at $+0.94$.

R^2 remains modest. Across rate specifications $R^2 \in [0.45, 0.50]$, meaning slightly more than half of the cross-country variation in per-cohort athlete supply is *unexplained* by our macro covariates. We attribute this residual primarily to country-specific recruitment pipelines: the Kenyan-distance-running ecosystem centered on Iten, the MVP Track Club and Racers Track Club networks that dominate Jamaican sprinting, similar conduits in the Bahamas and Trinidad, the Norwegian and Australian middle-distance scenes, and so on. These are pipeline structures that macro variables cannot capture and probably cannot proxy. A complete model of international NCAA supply would require sport-level network data we do not have.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26. Macro covariates are 2023 plus 2010–23 aggregates. A descriptive pseudo-panel constructed from athletes’ class-year (Fr/So/Jr/Sr/Gr) gives at most five entry cohorts. A static difference-in-differences with country and year fixed effects estimates a small positive interaction between COVID-19 stringency and the 2021–22 entry cohorts ($\hat{\beta} = +0.07$, $p = 0.06$, country-clustered SEs), consistent with the disaster-events story; we report this in `output/event_study.txt` as a descriptive exercise rather than as an identified event study, since the imputation-based construction has no clean pre-period and class-year mapping is noisy.
- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (school-page rendering issues that systematically affect schools with JS-rendered rosters, which skew toward a few major D-I programs).
- **Track-and-field dominance.** 96% of the international sample competes in track. The paper’s title and findings should be understood as describing the international track-and-field recruitment market specifically.

- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator for per-cohort rates. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals (only 5-year bands as percentages of gender-specific totals). Rescaling the 15–24 cohort by a constant fraction is observationally equivalent in the log-rate model, so we cannot directly test denominator sensitivity here; an extension using national census 18–22 counts would be valuable.
- **Athletes per capita is not athletic talent per capita.** Recruiting selection, scholarship caps, language and visa frictions, and U.S. schools’ bilateral relationships filter the population. We observe the *combined* effect of supply and selection.
- **Bartik weak instrument.** The shift-share construction on track-event composition has a first-stage $F = 13.5$, below the modern $F > 104.7$ threshold for valid 2SLS t -tests at the 5% level (?). We use it only as a covariate robustness check, not as an identification strategy.
- **Modest R^2 .** The residual 50–55% of cross-country variation in per-cohort supply is likely driven by country-specific recruitment pipelines that macro covariates cannot proxy.

6 Reproducibility

All results in this paper are reproducible from the local Python scripts in the project repository:

- `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`: build the roster data.
- `run_country_mapping.py`: resolves hometowns to country codes.
- `run_merge_analysis.py`: joins rosters to NCAA-IPEDS crosswalk, IPEDS HD, EADA, and macro indicators.
- `fetch_extra_shocks.py`, `run_join_shocks.py`: add the eight shock panels and population.
- `run_econ_distance.py`: adds the per-athlete economic-distance metrics.
- `run_analysis.py` and `run_robustness.py` produce the main coefficient tables.
- `run_stress_tests.py` produces the disaster-deaths drop-top-5, disaster-events alternative, refugees raw-count flip, and two-margin probit + intensity reported in this paper’s Tables 2 and 3 and Section ??.
- `run_bartik.py`, `run_event_study.py`, `run_paper_outputs.py`: build the auxiliary Bartik, COVID descriptive exercise, and Table 1 / figures.

Figures

References

Lee, D. S., J. McCrary, M. Moreira, and J. Porter (2022). “Valid t -Ratio Inference for IV.” *American Economic Review* 112(10): 3260–3290.

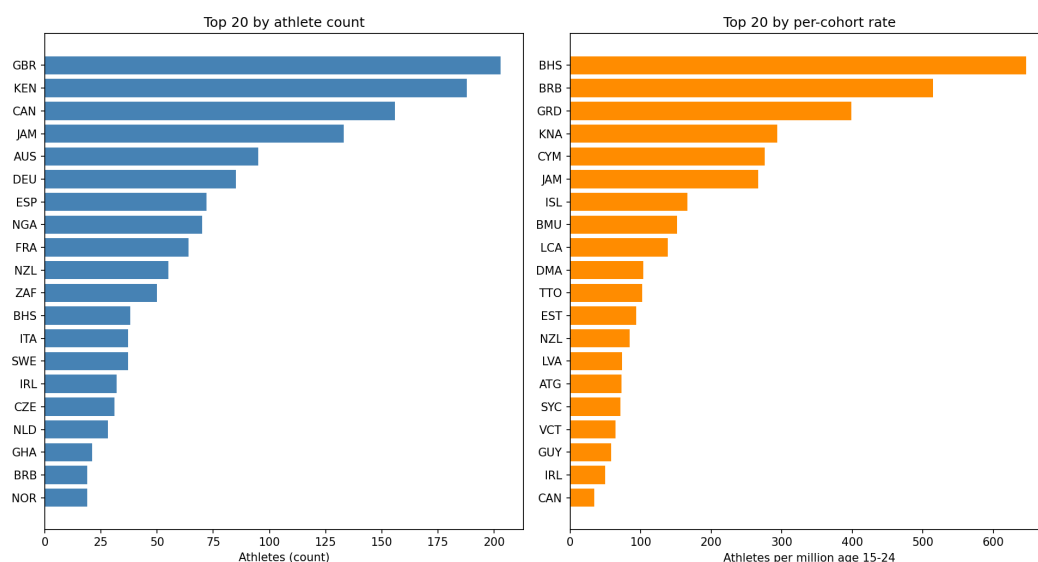


Figure 1: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

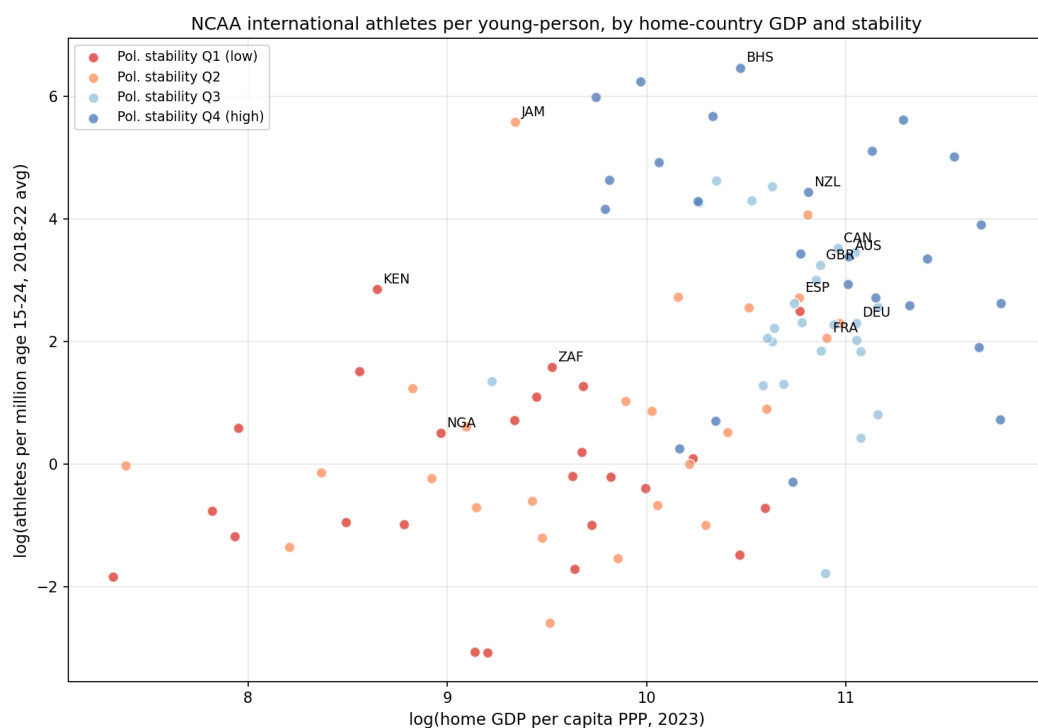


Figure 2: Log athletes per million age 15–24 vs. log home-country GDP per capita, color-coded by political-stability quartile. Top twelve senders annotated.

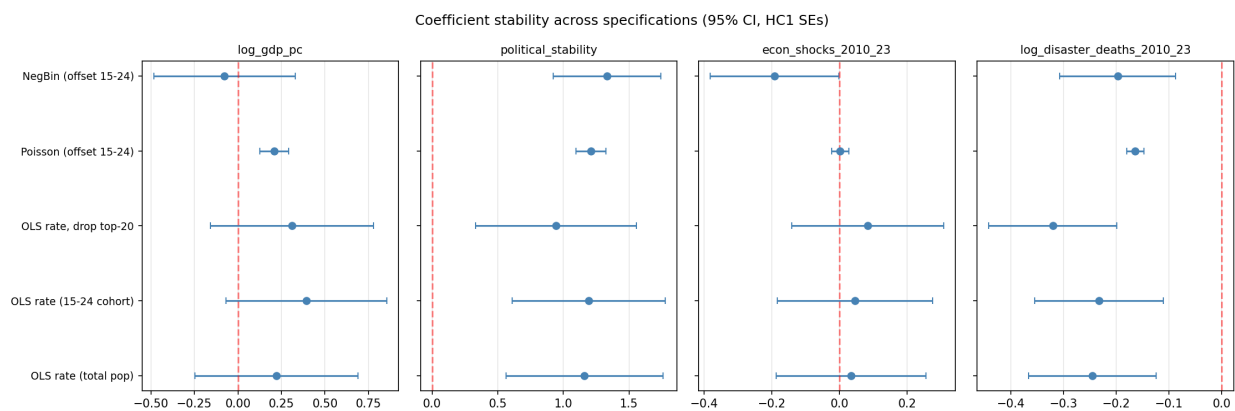


Figure 3: Coefficient stability across five specifications (OLS rate on total population, OLS rate on 15–24 cohort, OLS rate dropping top-20 senders, Poisson with cohort offset, Negative Binomial with cohort offset). 95% confidence intervals; HC1 robust SEs.